HOSPITAL OPERATIONS & CLINICAL STRATEGY DOSSIER

Phase II: SQL Relational Analytics & Flow Optimization

Project: NovaCare Health Systems – Operational Intelligence

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Status: Phase II Completion (Advanced SQL Integration)

EXECUTIVE SUMMARY

Following the successful sanitization of the core Health Performance Dataset in Excel (Phase I), this phase focuses on the relational integration of **Admissions**, **Diagnoses**, and **Patient** tables. By transitioning to a SQL-based environment, we have moved beyond simple data cleaning into **Clinical Throughput Modeling**. This report identifies systemic bottlenecks in discharge planning, department-level efficiency discrepancies, and the "Sunday Surge" phenomenon that impacts hospital-wide capacity.

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SECTION 1: DATA ARCHITECTURE & RELATIONAL SCHEMA

In Phase I, the data existed as isolated "flat" files. While useful for basic filtering, this format lacked the ability to track a patient's full journey across different clinical encounters. Phase II involved migrating this data into a **PostgreSQL** environment to establish a robust relational framework.

1.1 Structural Foundation: From Flat Files to Relational Tables

The transition from Excel to SQL moved the project from **Static Data** to **Dynamic Intelligence**. In the flat-file model, repeating patient information across every row created data redundancy and increased the risk of manual entry errors.

By defining a formal schema, we established three core tables:

1. **patient.csv**: The "Identity" table (Demographics, Age, Gender).
2. **admissions.csv**: The "Operational" table (Arrivals, Departments, Discharges).
3. **diagnoses.csv**: The "Clinical" table (ICD-10 coding and medical conditions).

This structure allows us to perform complex **JOIN** operations, enabling us to link a patient's demographic profile to their specific medical diagnosis and hospital stay duration simultaneously.

1.2 Schema Engineering: Primary Anchors

To ensure every record is unique and traceable, we implemented **Primary Anchors** (Unique Identifiers). This is the "glue" that holds the NovaCare database together.

- **admission_id**: Assigned to the admissions table as a unique key. This allows us to link multiple diagnoses to a single hospital stay without duplicating the stay information itself.
- **patient_id**: Acts as a **Foreign Key** in the admissions table. This ensures that even if a patient visits the hospital ten times, their demographic data (like age) is only stored once but referenced many times.
- **diagnosis_id**: Created in the diagnoses table to audit each specific medical condition flagged during an admission.

Technical Benefit: These anchors allow for **Relational Integrity**, meaning we can now audit "lost" records or revenue leakage by identifying admissions that lack an associated diagnosis code.

1.3 Data Purity Protocol: "True Pediatrics" vs. "Adult Overflow"

One of the most significant operational discoveries was the "contamination" of the Pediatric Ward. Initial queries showed patients as old as 65 being "admitted" to Pediatrics. To provide the hospital board with accurate specialty KPIs, we developed a **Data Purity Protocol** using SQL logic.

We applied a CASE statement to re-classify every record based on clinical reality rather than administrative labeling:

SQL

```sql
CASE
    -- Rule 1: Age-based classification
    WHEN p.age <= 17 THEN 'True Pediatrics'

    -- Rule 2: Identifying "Boarders" (Adults staying in the kids' ward)
    WHEN p.age >= 18 AND a.department = 'Pediatrics' THEN 'Adult Overflow'

    -- Rule 3: Preserving original specialty data for adults
    ELSE a.department
END AS final_department_mapping
```

Outcome: This protocol revealed that the "Pediatrics" department volume was artificially inflated by adult overflow. By sanitizing this column, we provided NovaCare with the first accurate look at their true pediatric resource utilization, which is essential for pediatric-specific staffing and budgeting.

SECTION 2: ADMISSION DYNAMICS & CAPACITY TRENDS

Understanding the rhythm of patient arrivals is essential for hospital staffing and resource management. By analyzing the admission_datetime across the 5,000-record dataset, we identified clear patterns that distinguish routine operations from high-pressure surges.

2.1 Daily Inflow Analysis: Baseline vs. Surges

Our SQL analysis shows that NovaCare Hospital maintains a **consistent baseline demand**, typically processing between **10 and 18 admissions per day**. This represents the "steady state" of the facility.

- **Intermittent Surges:** The data reveals frequent "spike days" where admissions exceed **20+ patients**.
- **Operational Risk:** These surges are significant because they push the triage and nursing staff beyond their standard capacity. When inflow exceeds the baseline by more than **25%**, the hospital transitions from "routine care" to "surge management," often leading to longer wait times and increased provider fatigue.

2.2 The Sunday Surge: The 737-Patient Peak

The most striking finding in the weekly distribution is the **Sunday Surge**. With **737 total admissions**, Sunday is the highest-volume day of the week, while Saturday (697) remains the lowest.

Statistical Breakdown:

- **The Weekend Backlog:** The 5.7% increase from Saturday to Sunday suggests a "wait-and-see" behavior by patients. Many individuals likely experience symptoms on Friday or Saturday but delay seeking hospital care until the end of the weekend.
- **Primary Care Gap:** The surge is also attributed to the closure of outpatient clinics and primary care offices on Sundays, leaving the NovaCare Emergency Department as the only available point of care.

2.3 Chronological Bottlenecks: The 5:00 AM Peak

A deep dive into the hourly admission data revealed a counter-intuitive peak at **5:00 AM (259 patients)**, making it the busiest hour for formal admissions.

The "Administrative Backlog" Investigation: In a typical clinical setting, emergency arrivals usually peak in the late afternoon or evening. The 5:00 AM spike in the data is identified as an **Administrative Bottleneck**:

- **Shift Transition Processing:** Patients arriving during the high-volume late-night hours (11 PM – 3 AM) are often stabilized first, but their formal "admission" paperwork is not completed until the morning shift transition begins.
- **The Queue Effect:** This "documentation dump" creates a false appearance of a morning surge.

- **Operational Impact:** This causes a massive workload for the incoming 6:00 AM shift, as they must manage a backlog of "newly admitted" patients who have actually been in the building for hours.

Strategic Insight: To resolve this, NovaCare should consider "Evening Documentation Clerks" to finalize admissions in real-time between midnight and 3:00 AM, flattening the 5:00 AM spike and allowing the morning clinical team to focus on direct patient care.

SECTION 3: CLINICAL THROUGHPUT (LENGTH OF STAY)

The Length of Stay (LOS) is a vital metric for NovaCare Hospital, representing the balance between clinical recovery and bed turnover. Our analysis reveals that while the hospital maintains a steady flow, the lack of variation across different patient profiles suggests that operational policies may be overriding individual clinical needs.

3.1 The Efficiency Paradox: Constant 4.0 Day LOS

A critical discovery in our SQL analysis is that the average LOS remains remarkably flat, hovering around **4.0 days** regardless of the patient's condition, age, or diagnosis. This "Efficiency Paradox" is unusual in a clinical setting where a minor stomach flu (A09) should theoretically result in a much shorter stay than a complex case of Pneumonia (J18).

- **Standardized Policy vs. Clinical Need:** The data indicates that LOS is likely driven by **standardized hospital discharge targets** or insurance-driven "expected stay" policies rather than patient recovery speed.
- **Operational Concern:** This flat trend raises the risk of "Premature Discharge" for highly complex patients or "Excessive Retention" for stable patients simply to meet a 4-day administrative window.

3.2 Departmental Variance: Surgery, Orthopedics, and ICU

When we segment the data by department, we see subtle but significant differences that reflect the intensity of care required.

- **Surgery & Orthopedics:** These departments exhibit the highest average LOS (4.1 to 4.2 days). This is expected, as post-operative recovery, wound management, and physical therapy assessments inherently require more time before a patient is safe for discharge.
- **ICU (Intensive Care Unit):** Despite treating the most critical patients, the ICU shows a slightly shorter stay (3.9 days). This often indicates a high-efficiency "stabilize and transfer" model, where patients are moved to general medical wards as soon as they are stable, freeing up critical care beds for new emergencies.
- **OB/GYN:** Maintains the fastest turnover, likely due to the highly standardized care pathways associated with "Normal Deliveries" (O80).

3.3 Admission Type Impact: Urgent vs. Emergency

Contrary to expectations, **Urgent admissions** often result in longer hospital stays than **Emergency admissions**.

- **Emergency (55.1% of volume):** While critical, these cases often have clear, aggressive treatment protocols that lead to faster stabilization or, unfortunately, quicker terminal outcomes.
- **Urgent (15.3% of volume):** Patients admitted under this category often present with complex, chronic issues that are not immediately life-threatening but require extensive diagnostic workups, specialty consultations, and stabilization before discharge.
- **Capacity Pressure:** Because Urgent patients occupy beds for longer durations on average, even a small increase in their volume has a disproportionate effect on the hospital's overall bed vacancy rate.

SECTION 4: DIAGNOSTIC MAPPING & AGE STRATIFICATION

By merging clinical codes with demographic data, we move beyond operational numbers and into the actual health profile of the NovaCare patient population. This section breaks down the "Why" and "Who" behind the admission trends.

4.1 Clinical Taxonomy: Decoding the ICD-10 Library

To make the data actionable for non-clinical stakeholders, we translated abstract alphanumeric codes into human-readable conditions. This mapping revealed a diverse case mix centered around acute and chronic management.

- **High-Volume Conditions:** The most frequent adult admissions are driven by **UTIs (N39)** and **Type 2 Diabetes (E11)**. These conditions require consistent monitoring and stabilization, often contributing to the steady 4-day stay.
- **Acute Interventions: Pneumonia (J18)** and **Acute Appendicitis (K35)** represent the primary drivers for the Surgery and Medicine departments, requiring high-intensity resources immediately upon admission.
- **Specialized Care: Normal Deliveries (O80)** dominate the OB/GYN ward, providing a predictable, high-volume flow that supports the hospital’s operational baseline.

4.2 Demographic Flow: Care Pathways by Age Group

The patient population was stratified into four distinct life stages to identify specific care needs and potential resource strains.

- **Pediatrics (0-17):** Shows high volume but specialized needs. Our SQL re-mapping was crucial here to ensure adult overflow did not mask the specific resource requirements for true pediatric patients.
- **Young & Middle Adults (18-64):** This group is the primary driver of Emergency admissions, often presenting with acute infections or injury-related Orthopedic cases.
- **Seniors (65+):** While this group does not necessarily stay significantly longer in the hospital, their **Care Pathways** are the most complex. Seniors are the primary population utilizing the **Rehab** and **Skilled Nursing** discharge tracks, making their throughput dependent on external facility availability.

4.3 Mortality & Outcome Correlation: The 4.35-Day Marker

A deeper look at the "Outcome" data revealed a sobering but important statistical trend regarding the 5,000 encounters.

- **The Mortality/LOS Connection:** Patients with an "Expired" outcome had a slightly higher average stay of **4.35 days**, compared to the 4.0-day average for those discharged home.
- **Critical Timing:** The small gap (only ~0.35 days) suggests that many critical patients are arriving at the hospital in a late-stage crisis. This "Limited Escalation Time" means the clinical team has a very narrow window to intervene before a terminal outcome occurs.
- **Resource Intensity:** These cases often involve high-utilization ICU resources in their final days, highlighting the need for early palliative care discussions or more aggressive triage for high-risk seniors arriving during "Surge" periods.

SECTION 5: THE DISCHARGE TRANSITION GAP (THE CORE BOTTLENECK)

The most significant finding of this operational audit is that NovaCare's "clogged" patient flow is not primarily a medical treatment issue, but a transition issue. While the clinical team is efficient at stabilizing patients, the system fails during the hand-off to post-hospital care.

5.1 Post-Acute Dependency: The 26% Pivot

Our SQL analysis of discharge dispositions reveals that while 71% of patients return home, a substantial **26% of the total patient volume** requires a transfer to a specialized recovery environment.

- **Rehab vs. Skilled Nursing:** Data shows a higher volume of referrals to **Rehabilitation Centers (784 patients)** compared to **Skilled Nursing Facilities (519 patients)**.
- **The "Active Recovery" Bias:** This suggests the NovaCare population is predominantly composed of patients who are expected to recover function (such as post-orthopedic surgery or stroke recovery) rather than those requiring long-term palliative or stationary nursing care.

5.2 External Resource Constraints: Recovery vs. Vacancy

At NovaCare, **clinical recovery does not equal an available bed.** This is the primary reason for the stagnant 4.0-day Length of Stay.

- **The Third-Party Variable:** Because 26% of discharges depend on external facility availability, a patient may be medically cleared for discharge on Day 3, but forced to remain in a hospital bed until Day 5 because the receiving Rehab center has no vacancy.

- **The "Holding Pattern":** These patients continue to occupy high-cost acute care beds, effectively "locking" the room and preventing new Emergency Department patients from being admitted. This creates the "boarding" crisis seen during peak surge hours.

5.3 Transition of Care: The Senior Discharge Delay

The "Discharge Gap" disproportionately affects the **Senior (65+) population**, who are the primary users of the Rehab and Skilled Nursing pathways.

- **Coordination Complexity:** Discharging a senior patient involves a multi-layered coordination process involving social workers, insurance adjusters (NovaCare's insurance arm), and the family.
- **The Logistics Trap:** Our data indicates that delays are often caused by the "Skilled Nursing" transition. Unlike home discharges, which can happen as soon as a ride is available, these transitions require specialized medical transport and pre-authorized insurance paperwork.
- **Systemic Impact:** By identifying this bottleneck, NovaCare can now pivot resources toward **Early Discharge Planning**—starting the paperwork for Senior transitions on the day of admission rather than the day of recovery—to shave critical hours off the total stay duration.

SECTION 6: FINAL RECOMMENDATIONS & STRATEGIC ROADMAP

The culmination of this SQL-driven analysis provides NovaCare Hospital with a data-backed blueprint for operational excellence. To move from "reactive" to "proactive" management, the following strategic steps are recommended for the final phase of the project.

6.1 Operational Alignment: Staffing for the 5:00 AM Surge

The discovery of the 5:00 AM administrative peak reveals a significant misalignment between current staffing schedules and the actual data-processing load.

- **Front-Loading Administration:** It is recommended that NovaCare implement an "Early-Response" administrative shift starting at 4:00 AM. By processing the overnight backlog before the 7:00 AM clinical shift change, the hospital can ensure that medical staff hit the floor with a clear, updated patient census.
- **Real-Time Processing:** To eliminate the "Documentation Dump," triage clerks should be incentivized to finalize formal admissions within two hours of arrival, rather than waiting for morning shift transitions. This will provide a more accurate, real-time picture of bed occupancy.

6.2 Predictive Planning: Data-Driven Resource Allocation

With the identification of the "Sunday Surge" (737 admissions), the hospital can now move away from static staffing models toward a dynamic, predictive approach.

- **The Sunday-Monday Bridge:** Because Sunday is the high-volume entry point, Monday is typically the day where the "Discharge Gap" is most felt as beds become full. Resource allocation for Social Workers and Discharge Coordinators should be increased on Monday mornings to clear beds for the Sunday arrivals.

- **Saturation Alerts:** Leadership should implement "Capacity Alerts" on Saturday evenings, preparing the facility for the statistically certain influx of patients on Sunday morning. This includes ensuring environmental services (cleaning) and pharmacy staffing are at peak levels to support faster turnover.

6.3 Future State: The Power BI Integration

The SQL queries developed in this phase serve as the "Logic Engine" for the upcoming **Phase III: Visualization & Executive Dashboarding**.

- **Dynamic Dashboards:** The ICD-10 mapping and LOS calculations will be fed into Power BI to create real-time heat maps of department saturation. This allows administrators to see exactly which wards (e.g., Orthopedics or Surgery) are nearing capacity at any given moment.
- **KPI Tracking:** Phase III will transform the "Clean_Department" SQL logic into visual gauges that track "True Pediatrics" volume versus "Adult Overflow," ensuring the hospital board has an undistorted view of facility performance.
- **Predictive Modeling:** By linking the historical Excel data with these new relational SQL insights, we will build a forecasting tool within Power BI that predicts next week's admission volume based on current day-of-week trends.

SECTION 7: CONCLUSION

The transition from Phase I’s data sanitization to Phase II’s relational SQL modeling has provided a definitive diagnostic of NovaCare Hospital’s operational health. We have successfully moved beyond looking at "what" happened to understanding "why" it happens.

The analysis proves that while NovaCare operates with high clinical efficiency—maintaining a steady 4.0-day stay across diverse diagnoses—it is currently limited by systemic administrative bottlenecks and external dependency. The identification of the **5:00 AM surge** and the **26% post-acute discharge gap** provides leadership with clear, high-impact targets for improvement. By implementing the recommended staffing realignments and proactive discharge planning, NovaCare can significantly reduce bed "boarding" times, increase daily capacity, and ensure that the hospital remains a high-performance integrated delivery network.

This SQL architecture now stands ready as the backbone for Phase III, where these insights will be transformed into live, visual intelligence to guide daily executive decision-making.

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